



FLOOD FORECASTING



*USING MACHINE
LEARNING*

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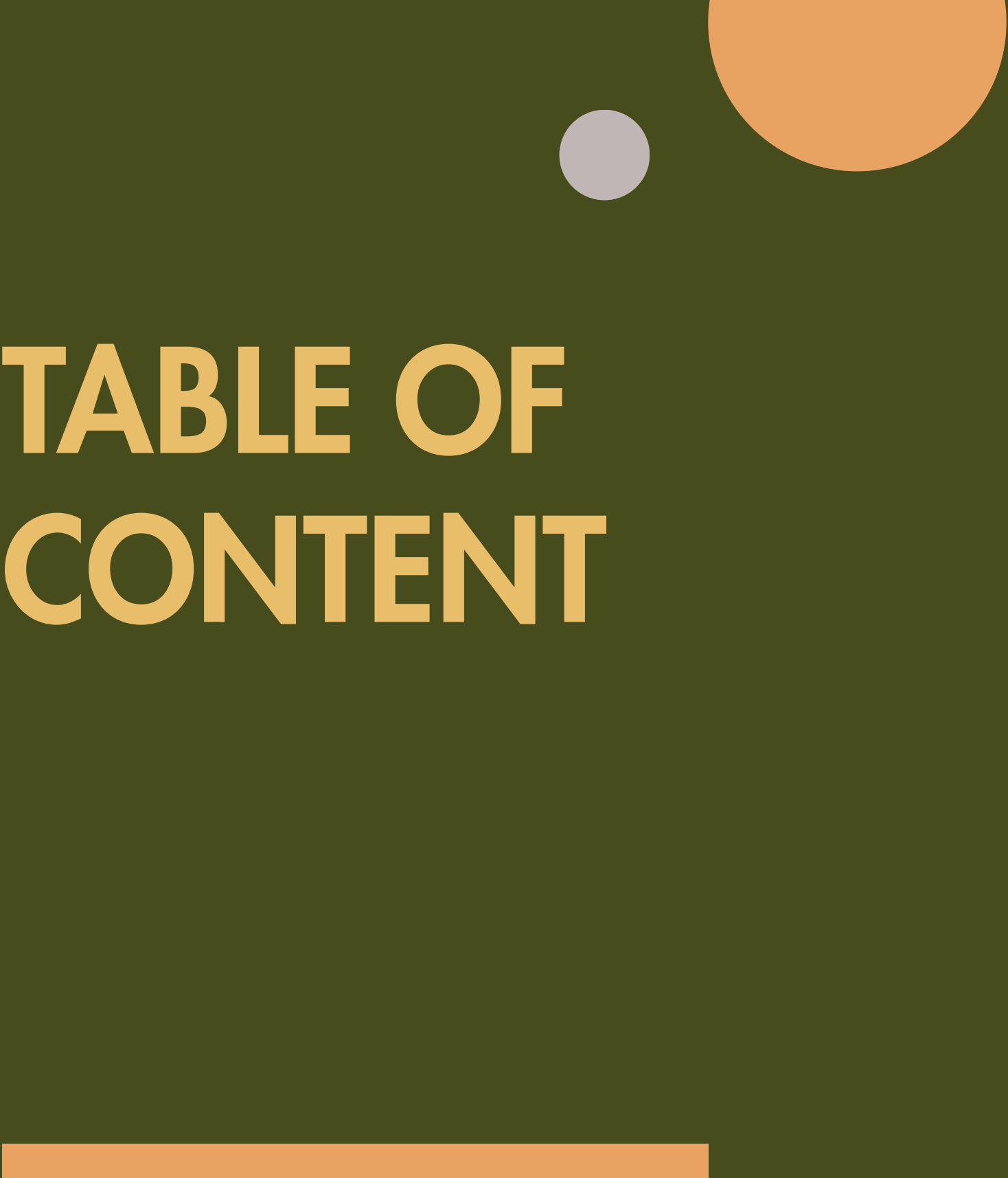


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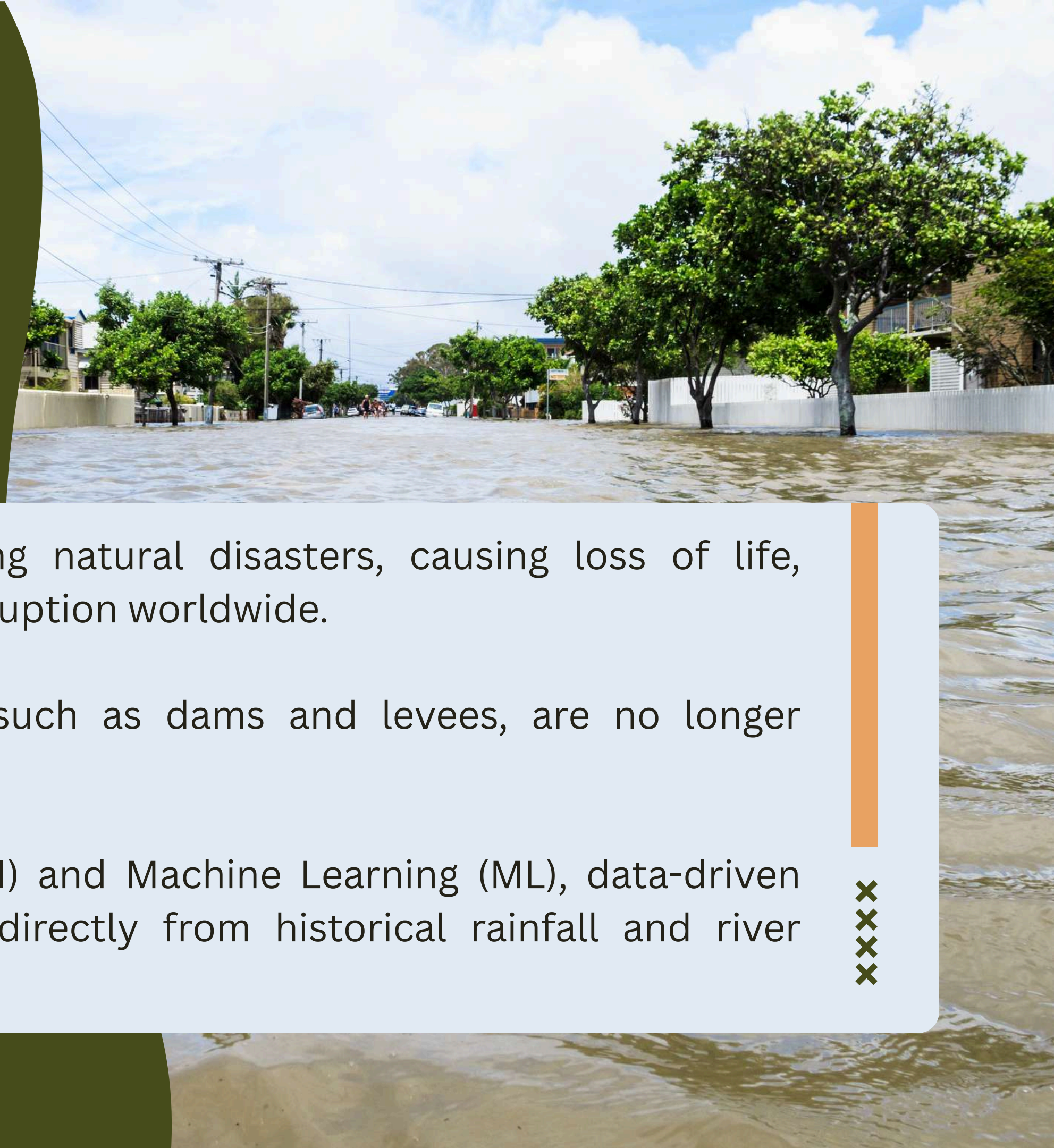
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INTRODUCTION

UNDERSTANDING THE NEED FOR FLOOD PREVENTION

- Floods are among the most devastating natural disasters, causing loss of life, infrastructure damage, and economic disruption worldwide.
- Traditional flood prevention methods, such as dams and levees, are no longer sufficient on their own.
- With the rise of Artificial Intelligence (AI) and Machine Learning (ML), data-driven models can now learn flood patterns directly from historical rainfall and river discharge data.



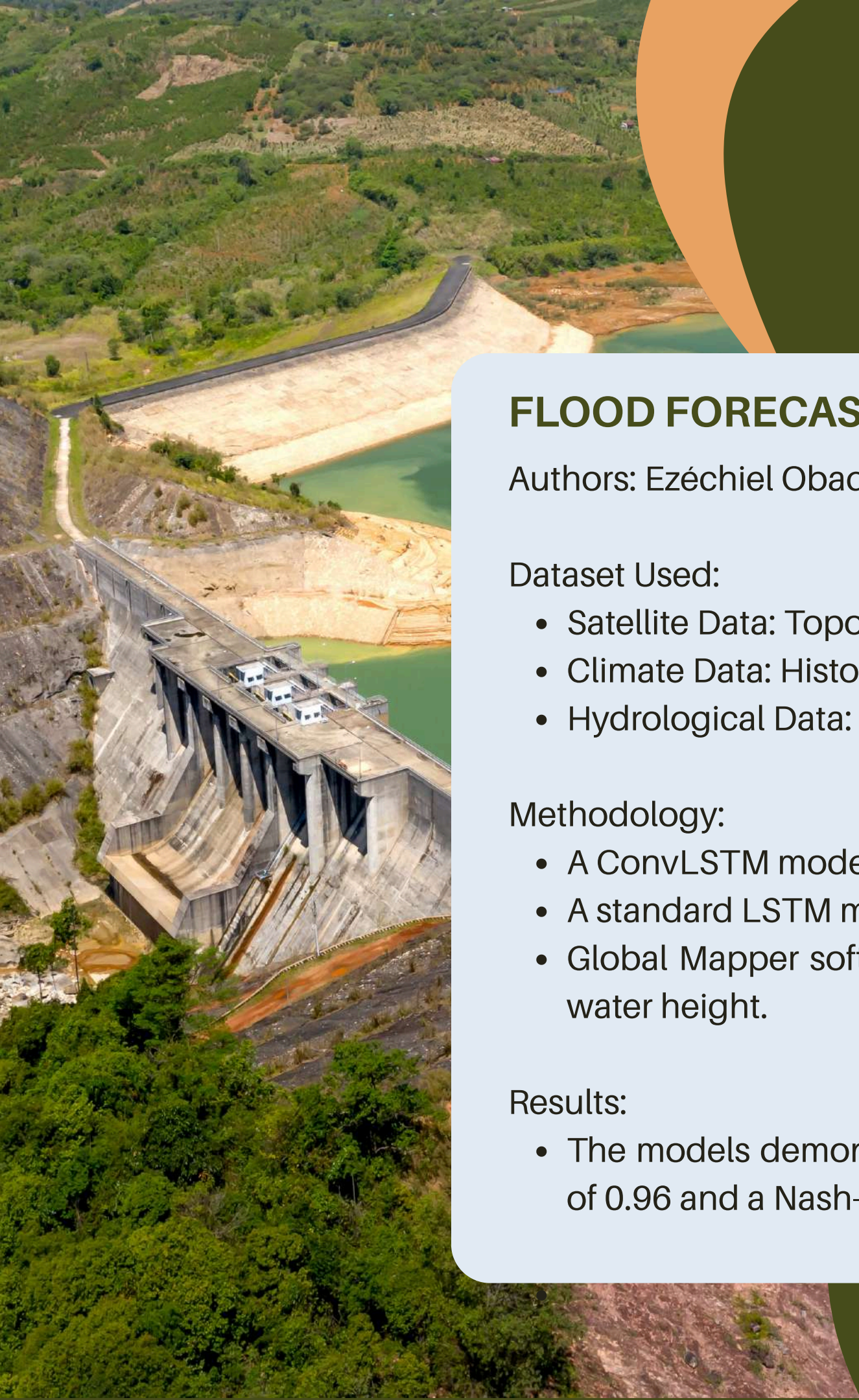
CAUSES OF FLOODING

Floods occur due to a combination of natural and human-induced factors, often leading to devastating consequences. Understanding these causes is essential for developing effective flood prevention strategies.

- **Heavy Rainfall & Storms**
- **River Channel Overflow**
- **Upstream Runoff**
- **Deforestation And Climate Change**

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LITERATURE REVIEW

FLOOD FORECASTING: THE CASE OF THE OUÉMÉ BASIN AT THE BÉTÉROU OUTLET

Authors: Ezéchiél Obada; E. I. Biao; P. J. Zohou; H. Yarou; F. Z. Hounnondaho; E. A. Alamou

Dataset Used:

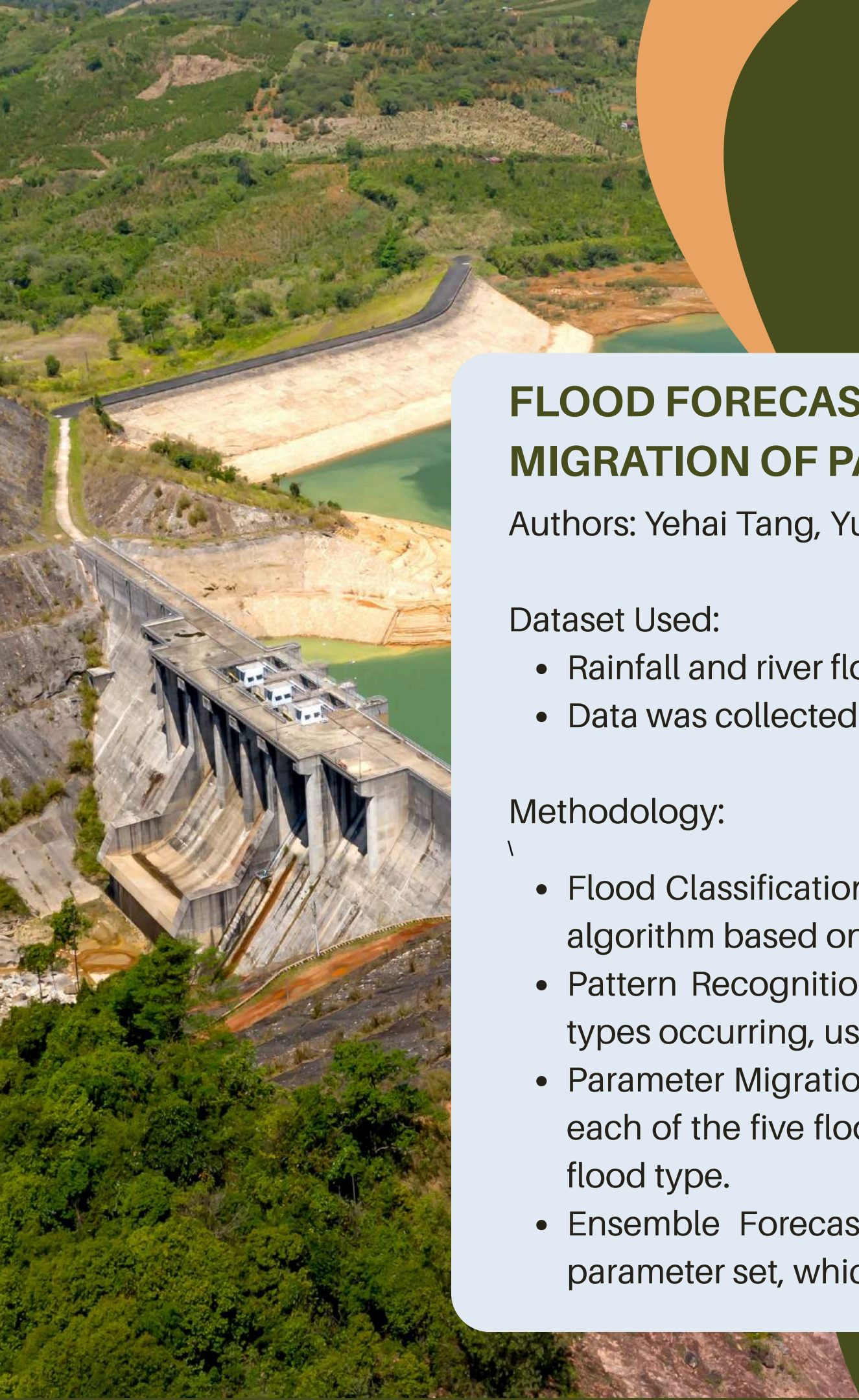
- Satellite Data: Topographical imagery from the US Geological Survey (USGS).
- Climate Data: Historical precipitation, temperature, and wind speed from the ERA5 database.
- Hydrological Data: In-situ river flow measurements and water height data.

Methodology:

- A ConvLSTM model was used to process spatio-temporal climate data and forecast river discharge.
- A standard LSTM model then predicted the corresponding water height from the forecasted discharge.
- Global Mapper software was used to simulate and visualize the flood inundation area based on the predicted water height.

Results:

- The models demonstrated high performance, with the ConvLSTM achieving a coefficient of determination (R^2) of 0.96 and a Nash-Sutcliffe efficiency (NSE) of 0.95.



LITERATURE REVIEW

FLOOD FORECASTING BASED ON ML PATTERN RECOGNITION AND DYNAMIC MIGRATION OF PARAMETERS

Authors: Yehai Tang, Yue Sun, Zhenyue Han, Shan-e-hyder Soomro, Qiang Wu, Boyu Tan, Caihong Hu

Dataset Used:

- Rainfall and river flow data from 98 flood events (1971-2014) in the Jingle sub-basin of the Yellow River, China.
- Data was collected from 13 rainfall stations and 1 hydrological station and processed into 1-hour time steps.

Methodology:

- Flood Classification: Historical floods were first categorized into five distinct types using the k-means clustering algorithm based on their characteristics (e.g., peak discharge, duration).
- Pattern Recognition: A Random Forest model was trained to predict the probability of each of the five flood types occurring, using only rainfall characteristics as input.
- Parameter Migration: An optimal set of parameters for the Xin'anjiang hydrological model was pre-calibrated for each of the five flood types. For a new event, the model selects the parameters corresponding to the predicted flood type.
- Ensemble Forecasting: The final prediction is a probability-weighted average of the forecasts from each parameter set, which improves accuracy and reduces uncertainty.



PROBLEM STATEMENT



Predict river water level or discharge in advance to minimize flood damage using publicly available hydrological data.



PROPOSED SOLUTION

Step 1: Data Collection

- Gather historical data such as rainfall, river discharge, temperature, soil moisture, and water level from reliable sources (meteorological and hydrological departments).

Step 2: Data Preprocessing

- Handle missing values, remove outliers, normalize data, and align time-series records for consistency.

Step 3: Feature Selection

- Identify key parameters influencing floods — rainfall intensity, upstream discharge, soil saturation, etc.

Step 4: Model Development

- Apply machine learning and deep learning algorithms (e.g., Random Forest, LSTM, GRU) to learn flood patterns from past data.

Step 5: Model Training and Validation

- Split dataset into training and testing subsets to evaluate prediction accuracy using metrics like RMSE, MAE, and R^2 .

Step 6: Flood Prediction

- Use the trained model to forecast future water levels or flood likelihood at specific locations or time intervals through dashboards or maps and trigger early warnings when predicted values exceed threshold limits.





OBJECTIVES

- 1.To collect and preprocess relevant hydrological and meteorological data for accurate analysis.
- 2.To identify key parameters influencing flood occurrence using statistical and correlation techniques.
- 3.To develop machine learning and deep learning models for predicting river water levels or flood events.
- 4.To evaluate model performance using standard accuracy and error metrics.
- 5.To design a simple visualization or alert system for early flood warning.
- 6.To demonstrate how AI-based models can complement traditional hydrological forecasting methods.



MODEL PERFORMANCE COMPARISON

Model	RMSE (cfs)	Relative Performance
Persistence (Baseline)	64,447	Reference
Random Forest	66,003	Slightly worse
Gradient Boosting	61,764	Best (+4.2%)

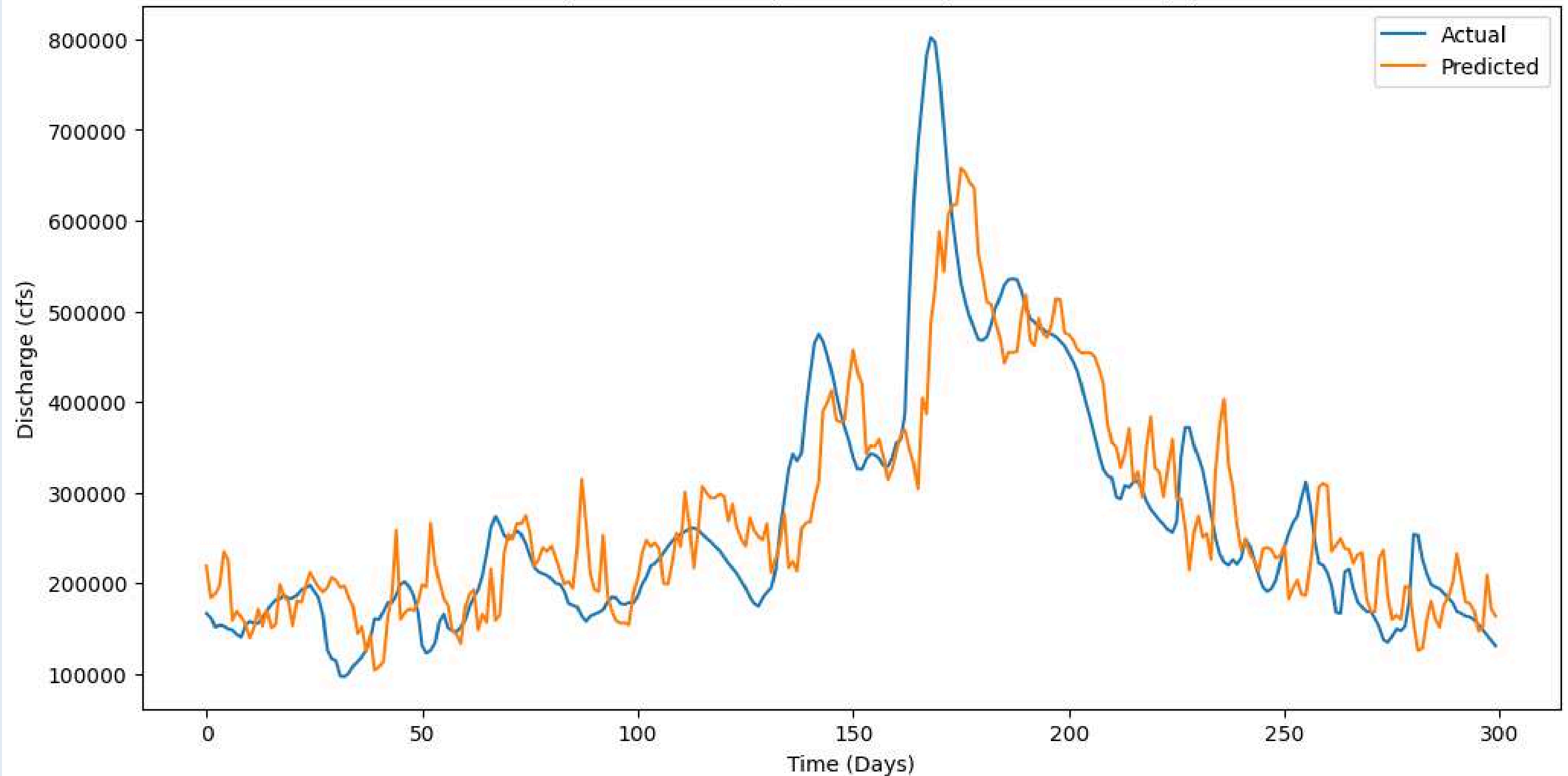


GRADIENT BOOSTING IMPROVED FORECASTING ACCURACY BY APPROXIMATELY 4-5% OVER THE PERSISTENCE BASELINE.

THE STRONG PERFORMANCE OF THE PERSISTENCE BASELINE REFLECTS HIGH TEMPORAL AUTOCORRELATION IN RIVER DISCHARGE.

FORECAST VISUALIZATION

7-Day Ahead Discharge Prediction (First 300 Test Days)

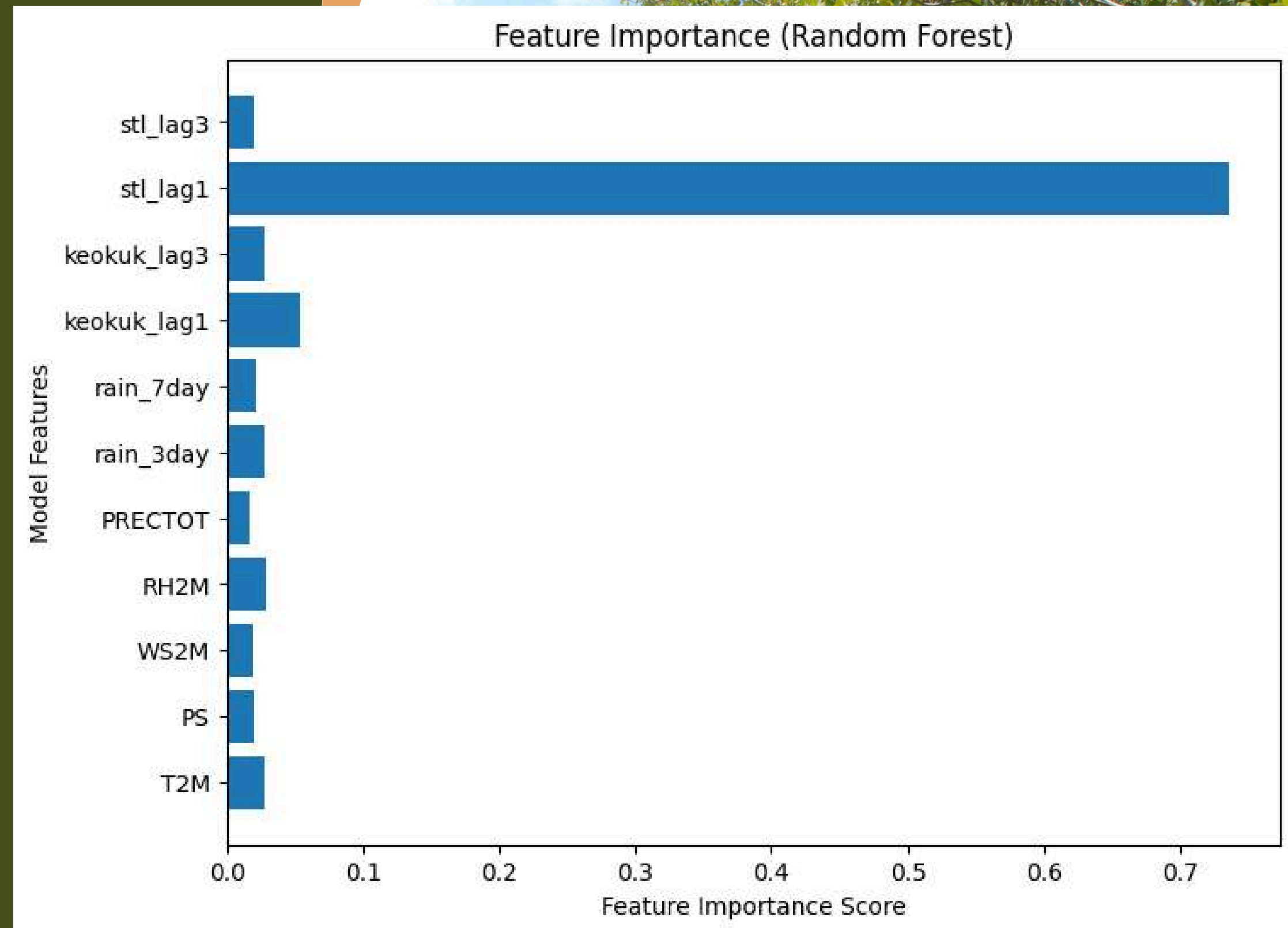


THE MODEL CAPTURES OVERALL DISCHARGE TRENDS AND MAJOR FLOOD EVENTS BUT SLIGHTLY UNDERESTIMATES EXTREME PEAKS.

FEATURE IMPORTANCE

- stl_lag1 dominates (~74%)
- Upstream flow (Keokuk) is second most important
- Rainfall variables contribute but are secondary

Short-term discharge forecasting is strongly driven by hydraulic persistence and upstream propagation.



REFERENCES



- Yaseen, Z.M., et al. (2019). "River Flow Forecasting Using Hybrid Artificial Intelligence Models." *Environmental Modelling & Software*, 122, 104521.
- Mosavi, A., Ozturk, P., & Chau, K.W. (2018). "Flood Prediction Using Machine Learning Models: Literature Review." *Water*, 10(11), 1536.





THANK YOU.

